

Individual Reflection: Data Science Toolbox, Formative Sprint 0

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Fit

Our group chose sports analytics. Each of us took a different dataset and found the resources that exist for analysing it. I worked on Premier League football using the Python package `penaltyblog` (Eastwood, `penaltyblog` documentation).

I took the perspective of a sports betting company, so my question was how a bookmaker can use a statistical model to price match outcomes, and how accurate and profitable those prices are. I started from the package's end-to-end example, which downloads three seasons of results and odds from `football-data.co.uk`, fits a Dixon-Coles model and predicts one fixture across several markets. I then produced four visualisations of my own using the same three seasons data from the example so I could check my results against the documentation.

Depth

Results: Bookmaker margins vary, the median margin for Bet365 was about 5.4% but only 2.8% for Pinnacle, with the market average around 4%. Customers lose about 7% of their bet on average, and long shots with odds of 10 or more lose about 37%, which is consistent with the favourite/longshot bias. My conclusion is that a bookmaker's margin is not spread evenly, and that even a one goal model can price many markets consistently.

Mathematics: Dixon-Coles models the goals scored by the home team (X) and away team (Y) when team i hosts team j as Poisson, with rates depending on attack strength α , a defence parameter δ (lower means fewer goals conceded) and a home advantage γ :

$$\lambda_{ij} = \exp(\gamma + \alpha_i + \delta_j), \quad \mu_{ij} = \exp(\alpha_j + \delta_i).$$

A plain Poisson model assumes X and Y are independent, which is inaccurate for low scores. Dixon and Coles (1997) correct this with a factor τ and a parameter ρ :

$$P(X = x, Y = y) = \tau_{\lambda, \mu}(x, y) \frac{e^{-\lambda} \lambda^x}{x!} \frac{e^{-\mu} \mu^y}{y!}, \quad \tau_{\lambda, \mu}(x, y) = \begin{cases} 1 - \lambda\mu\rho & x = y = 0, \\ 1 + \lambda\rho & x = 0, y = 1, \\ 1 + \mu\rho & x = 1, y = 0, \\ 1 - \rho & x = y = 1, \\ 1 & \text{otherwise.} \end{cases}$$

The parameters maximise a weighted log likelihood over matches k , where recent matches count more:

$$\ell = \sum_k \phi(t_k) \log P(X_k = x_k, Y_k = y_k), \quad \phi(t) = e^{-\xi(t_{\text{now}} - t)}.$$

Every market is a sum over the scoreline grid, for example $P(\text{home win}) = \sum_{x > y} P(x, y)$, which my heatmap displays. A bookmaker's margin is $\sum_k 1/o_k - 1$ for decimal odds o_k on the three outcomes.

Weaknesses: The model only uses goals, so it ignores injuries and other relevant features. The example fits and predicts on the same matches and never tests on unseen data, so I cannot say how accurate it is. My average return of about -7% is larger than the margin of about 4%, which I think is due to using only three seasons of data. Next time I would hold out the latest season and compare the model with the bookmakers' odds.

Literature search

My literature search was limited. I relied on the package documentation, which is written by the package's author and so presents it favourably, and I did not read the original Dixon and Coles paper in full.

My comfort zone is probability modelling, so understanding the mathematics behind the Dixon-Coles model was the easiest part. I struggled most with coding and learning the Python plotting packages. There are many ways to do the same thing in matplotlib and seaborn, and tasks like annotating points or formatting a heatmap took me a long time to understand. I had no technical issues with penaltyblog itself. To use these resources better I need more fluency in coding, especially plotting.

Collaboration

This was my first time using GitHub, and I found it difficult to learn how to work on it collaboratively, for example committing and pulling changes to my section. It did let us work in parallel in separate folders, but because we used different packages, we only compared approaches loosely.

Evidence

My contribution is the penaltyblog notebook (`01-penaltyblog-analysis.ipynb`), containing the downloaded example code and my four visualisations with analysis, together with the saved CSV of the data. It is in my folder and in the report folder.

References

Dixon, M. J. and Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 46(2), 265-280.

Eastwood, M. penaltyblog documentation. <https://penaltyblog.readthedocs.io>